Notes on algebraic combinatorics

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Abstract

This article consists of some notes taken by the author while studying algebraic combinatorics.

References: [1], [2], [3], [4].

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1 Some motivating results in combinatorics

We start by presenting two results which are special cases of the Heron-Rota-Welsh conjecture and the Dowling-Wilson conjecture.

1.1 The read conjecture

Given a graph G and a positive integer t, a **proper coloring** of G is an assignment of one of t-colors to each vertex of G such that the two endpoints of each edge are assigned different colors. The **chromatic polynomial** of G is defined to be the total number of proper colorings as a function of t, denoted by $P_G(t)$.

Example 1.1.

- *When G has a loop, $P_G(t) = 0$.*
- *When G is a connected tree with n-vertices, $P_G(t) = t(t-1)^{n-1}$.*
- *When G is the complete graph on n vertices, $P_G(t) = t(t-1)\cdots(t-n+1)$.*

The chromatic polynomial satisfies the following **deletion-contraction** relation.

Lemma 1.1 (deletion-contraction)**.** *Given any edge e of G, we have*

$$P_G(t) = P_{G\setminus e}(t) - P_{G/e}(t),$$

where $G \setminus e$ and G/e are the graphs obtained from G by deleting and contracting the edge e, respectively.

Using inductions on the number of edges, one can easily deduce the following properties from the deletion-contraction relation.

Corollary 1.1.

- *For any loopless graph G, the chromatic polynomial $P_G(t)$ is a monic polynomial.*
- *The coefficients of $P_G(t)$ are alternating, that is, if $P_G(t) = t^n + a_1 t^{n-1} + \cdots + a_n$, then $(-1)^i a_i \geq 0$.*

The following result of Huh confirms the Read conjecture from the 60's.

Theorem 1.1 (Huh, 2012)**.** *The (absolute values of the) coefficients of $P_G(t)$ form log-concave sequence. In other words, if $P_G(t) = t^n + a_1 t^{n-1} + \cdots + a_n$, then $|a_{i-1}a_{i+1}| \leq a_i^2$.*

Remark 1.1. *In fact, the sequence $|a_1|, \ldots, |a_n|$ has no internal zeros (which follows from Huh's proof, but is also a previously known result). Obviously, among the nonzero terms, $|a_{i-1}a_{i+1}| \leq a_i^2$ is equivalent to $\log |a_i|$ being concave. For the rest of the note, when we say a sequence a_k is log-concave, we mean the sequence $|a_k|$ has no internal zeros, and $|a_{k-1}a_{k+1}| \leq a_k^2$ for all k.*

1.2 The realizable Dowling-Wilson conjecture

Let V be a d-dimensional vector space over a field, and let $E \subset V$ be a finite generating set. Let

$$\mathcal{F} = \{\text{all linear subspaces of } V \text{ generated by a subset of } E\},$$

and $\mathcal{F}_k = \{F \in \mathcal{F} \mid \dim F = k\}$. Denote the cardinality of $\mathcal{F}_k$ by W_k. The sequence of numbers W_k are called **Whitney numbers of the second kind**. The following result confirms a conjecture of Dowling-Wilson in the realizable case.

Theorem 1.2. *Let W_k be defined as above. For $k \leq d/2$, we have $W_k \leq W_{d-k}$.*

It turns out that for both theorems, the proofs essentially use algebraic geometry. In fact, the first theorem reduces to the Hodge index theorem, and the second reduces to the hard Lefschetz theorem for projective varieties.

Both statements can be generalized to **matroids**, where the corresponding variety may not exist. Then the key idea is to produce some combinatorial proofs of the

analogous statements (Hodge index theorem, or more generally the Hodge-Riemann relations, and the hard Lefschetz theorems) on some combinatorial cohomology ring of matroids (or Chow rings, or intersection cohomology groups).

2 Introduction to matroids

Definition 2.1 (independent set)**.** *In graph theory, an independent set, stable set, coclique or anticlique is a set of vertices in a graph, no two of which are adjacent.*

Definition 2.2. *Given a finite set E in a vector space, we can consider the independent sets of E, defined as*

$$\mathcal{I}_E = \{I \subset E \mid I \text{ is an independent set}\}.$$

Definition 2.3. *Given a graph G with edge set $E(G)$, we can define the set of forests,*

$$\mathcal{I}_G = \{I \subset E(G) \mid I \text{ does not contain a cycle}\}.$$

The set of independent sets and forests share a common combinatorial property, called the exchange lemma.

Lemma 2.1 (Steinitz exchange lemma)**.** *Let U and W be finite subsets of a vector space V. If U is a set of linearly independent vectors, and W spans V, then:*

1. *$|U| \leq |W|$;*
2. *There is a set $W' \subseteq W$ with $|W'| = |W| - |U|$ such that $U \cup W'$ spans V.*

Lemma 2.2 (exchange lemma)**.** *Let $\mathcal{I}_E$ be defined as above. If $I_1, I_2 \in \mathcal{I}_E$ satisfies $|I_2| > |I_1|$, then there exists $x \in I_2$ such that $I_1 \cup x \in \mathcal{I}_E$*[1]*. The same statement holds for the set of forests $\mathcal{I}_G$.*

Matroid is the combinatorial structure that captures the independence conditions from both linear algebra and graphs.

Definition 2.4 (matroid)**.** *A **matroid** consists of a finite set E and a collection of subsets $\mathcal{I} \subset 2^E$, satisfying the following properties.*

1. *$\emptyset \in \mathcal{I}$;*
2. *if $I \in \mathcal{I}$ and $I' \subset I$, then $I' \in \mathcal{I}$;*
3. *if $I_1, I_2 \subset \mathcal{I}$ and $|I_1| < |I_2|$, then there exists $x \in I_2$ such that $I_1 \cup x \in \mathcal{I}$.*

*The sets in $\mathcal{I}$ are called **independent sets**.*

[1] Throughout this note, we abuse notations and use the element x to also denote the singleton set$\{x\}$.

When E is a finite subset of a vector space, the independent sets $\mathcal{I}_E$ defines a matroid.

There are many other equivalent definitions of matroids in terms of flats, rank functions, closure operators, bases, basis polytopes, etc. Here, we mention a few.

Definition 2.5. *A matroid is a pair $(E, \mathcal{B})$, where E is a finite set and $\mathcal{B} \subset 2^E$, satisfying the following properties.*

1. *$\mathcal{B}$ is nonempty;*
2. *If $A, B \in \mathcal{B}$ and $x \in A \setminus B$, then there is $y \in B \setminus A$ such that $(A \setminus x) \cup y \in \mathcal{B}$.*

*The sets in $\mathcal{B}$ are called **bases**.*

Definition 2.6. *A matroid is a pair $(E, \mathcal{F})$, where E is a finite set and $\mathcal{F} \subset 2^E$, satisfying the following properties.*

1. *$\mathcal{F}$ is nonempty;*
2. *if $F_1, F_2 \in \mathcal{F}$, then $F_1 \cap F_2 \in \mathcal{F}$;*
3. *given any $F \in \mathcal{F}$, every element of $E \setminus F$ is in exactly one minimal set in $\mathcal{F}$ which strictly contains F.*

*The sets in $\mathcal{F}$ are called **flats**.*

References

[1] Karim Adiprasito, June Huh, and Eric Katz. "Hodge theory for combinatorial geometries". In: *Annals of Mathematics* 188.2 (2018), pp. 381–452.

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[3] June Huh. "Milnor numbers of projective hypersurfaces and the chromatic polynomial of graphs". In: *Journal of the American Mathematical Society* 25.3 (2012), pp. 907–927.

[4] June Huh and Eric Katz. "Log-concavity of characteristic polynomials and the Bergman fan of matroids". In: *Mathematische Annalen* 354 (2012), pp. 1103–1116.